

Mashupstack Full Stack Course — Recap Notes

1. HTML

Structure

- `<!DOCTYPE html>` → `<html lang="en">` → `<head>` (meta, title, link, style) → `<body>`
- Semantic tags: `<h1>`–`<h6>`, `<p>`, `<blockquote>`, `<q>`, `<dl>`/`<dt>`/`<dd>`, `<ul>`/`<ol>`/`<li>`

Key Elements

Element	Attributes	Lab
<code></code>	src, alt, height, width	Lab1
<code><a></code>	href, target="_blank"	Lab1, Exam1
<code><table></code>	border, cellpadding, cellspacing, <code><tr></code> , <code><th></code> , <code><td></code>	Lab2, Exam1
<code><form></code>	action, target, onsubmit	Lab3, Exam1
<code><input></code>	type="text/email/number/submit/checkbox", required	Lab3
<code><iframe></code>	src, width, height, allowfullscreen	Lab2
<code><fieldset></code>	Groups form fields	Lab3

Exam1 Project — “Overlooked” (News Portal)

- Table-based layout for header, nav, content sections
 - Multi-page site with navigation, contact form, about page
 - YouTube embeds, comment forms on each news article
-

2. CSS

Selectors

Type	Example	Lab
Element	<code>div { }</code>	Lab5
Class	<code>.box1 { }</code>	Lab5
ID	<code>#heading { }</code>	Lab6
Universal	<code>* { }</code>	Exam2
Attribute	<code>input[type=email]</code>	Lab9
Descendant	<code>.container h1</code>	Lab8
Child	<code>li:first-child > a</code>	Lab8
Adjacent Sibling	<code>p + h3</code>	Lab8

Box Model

margin → border → padding → content

- border: style width color; border-radius
- margin: auto (center), max-width, min-width

Layout Systems

System	Properties	Lab
Flexbox	<code>display: flex; justify-content: center; align-items: center</code>	Lab6
CSS Grid	<code>display: grid; grid-template-columns; grid-template-rows</code>	Lab9
Float	<code>float: left/right</code>	Lab7, Lab8
Positioning	<code>position: static/relative/absolute/fixed</code>	Lab7, Exam2

Visual Effects

- `box-shadow, backdrop-filter: blur()` (glassmorphism)
- `background: linear-gradient(), background-size, background-attachment: fixed`
- `opacity, transition: opacity 0.2s`
- `border-radius: 50%` (circular images)

Pseudo-classes & Pseudo-elements

- `:hover, :active, :visited, :link, :first-child` → Lab6, Lab8
- `::first-line, ::first-letter, ::before` → Lab8

Responsive Design

- `@media` only screen and (max-width: 600px/768px)
- `@media` only screen and (orientation: landscape)
- Column hiding, font-size adjustments at breakpoints

Lab Progression

Lab	Topic
5	Box model, borders, padding, margin
6	Flexbox, tables, glassmorphism
7	Float, box-shadow, fixed positioning
8	Pseudo-elements, combinators
9	CSS Grid, media queries
10	Bootstrap 4 (navbar, grid)
Exam2	Mini projects: tables, buttons, login forms

3. Bootstrap

Grid System

```
<div class="container">
  <div class="row">
    <div class="col-lg-4 col-sm-12">...</div>
    <div class="col-lg-8 col-sm-12">...</div>
  </div>
</div>
```

- Breakpoints: `col-`, `col-sm-`, `col-md-`, `col-lg-`
- `.container-fluid` for full-width

Components Used

Component	Classes/Pattern	Lab
Navbar	.navbar .navbar-expand-lg .navbar-dark .bg-dark	Exam3, Lab14
Cards	.card .card-body .card-title .card-text .card-img-top	Exam3, Lab13
Tables	.table .table-striped .thead-dark	Lab11
Alerts	.alert .alert-success/warning/danger	Lab11
Buttons	.btn .btn-primary/success/outline-primary	All
Badges	.badge .badge-primary	Lab12
Progress Bars	.progress .progress-bar .progress-bar-striped .progress-bar-animated	Lab12
Forms	.form-group .form-control .form-check .form-range	Lab15
Modals	data-toggle="modal" data-target="#myModal"	Lab16
Accordion	#accordion .card .card-header data-toggle="collapse"	Lab13
Carousel	.carousel .carousel-inner .carousel-item .carousel-caption	Lab16
Dropdowns	.dropdown .dropdown-toggle .dropdown-menu .dropdown-item	Lab13
Pagination	.pagination .page-item .page-link	Lab12
Breadcrumbs	.breadcrumb .breadcrumb-item	Lab12
Tooltips	data-toggle="tooltip" title="..."	Lab16
Nav pills	.nav .nav-pills .nav-link.active	Lab14

Utilities

- Spacing: m-*, p-*, mx-auto, my-*
- Colors: .bg-primary/dark/light/success/warning, .text-white/dark/muted
- Display: .d-flex, .d-block, .w-100, .img-fluid, .rounded-circle, .shadow

4. JavaScript

Fundamentals

- **Variables:** var (function-scoped), let (block-scoped), const (constant reference)
- **Data Types:** number, string, boolean, object, array, undefined, NaN
- **Operators:** +, -, *, /, **, %, ===, !==, &&, ||, ternary (? :)
- **Type Conversion:** Number(), parseFloat(), parseInt(), String(), implicit coercion

Control Flow

```

if (condition) { } else if { } else { }
switch(value) { case 1: break; default: }
for (let i = 0; i < arr.length; i++) { }
while (condition) { }
for (let x in obj) { }

```

Functions

- Declaration: function name(params) { }
- Arrow: const fn = (params) => { }
- Callback: .filter(item => item.price > 50), .reduce((acc, item) => acc + item.score, 0)
- Recursion: function calling itself with modified parameters

Arrays

- `.push()`, `.filter()`, `.map()`, `.reduce()`, `.sort()`, `.join()`, `.length`
- `for` loop / `for...of` iteration

Objects

- Literal: `{ key: value, method() { } }`
- Access: `obj.key`, `obj["key"]`
- `in` operator, `delete` operator, `this` keyword

String Methods

- `.trim()`, `.toLowerCase()`, `.replace()`, `.split()`, `.indexOf()`, `.length`, `.match()`

DOM Manipulation

```
document.getElementById("id")
document.querySelector("selector")
element.innerHTML / .innerText / .style.property / .value
event.preventDefault()
```

Events

- `onclick`, `onmouseover`, `onmouseout`, `oninput`
- `window.addEventListener('load', function(){ })`

ES6+ Features

- **Template Literals:** ``Hello ${name}``
- **Arrow Functions:** `() => { }`
- **Classes:** `class X { constructor() { } get prop() { } set prop(v) { } }`
- **Spread:** `Math.max(...arr)`

Advanced

- **JSON:** `JSON.stringify()`, `JSON.parse()`
- **localStorage:** `setItem()`, `getItem()`
- **Error Handling:** `try { } catch(err) { }`
- **Regular Expressions:** `/pattern/flags`, `.test()`, `.match()`
- **Date:** `new Date()`, `.getFullYear()`, `.getMonth()`, `.getDate()`
- **setTimeout():** delay function execution

Lab Progression

Lab	Topic
18	Objects with methods, <code>this</code>
19	String manipulation
20	Math object, type coercion
21	Ternary, <code>in/delete</code>
22	Arrays, filter/reduce/sort
23	Date object, age calc
24	while loops, switch/case
25	Regular expressions
26	JSON, try/catch
27	ES6 classes, getters/setters
28	DOM events, <code>setTimeout</code>
Problems	Recursion, dark mode, anagrams

Lab	Topic
Exam	Calculator + lottery ticket

5. MySQL

DDL (Data Definition Language)

```
CREATE DATABASE name;
USE name;
CREATE TABLE name ( col DATATYPE CONSTRAINTS, ... );
ALTER TABLE name ADD column_name DATATYPE AFTER existing_col;
DROP DATABASE name;
DROP TABLE name;
TRUNCATE TABLE name;
```

Constraints

- PRIMARY KEY, AUTO_INCREMENT, NOT NULL, UNIQUE
- FOREIGN KEY (col) REFERENCES other_table(col)
- INDEX (col)

DML (Data Manipulation Language)

```
INSERT INTO table (cols) VALUES (vals);
SELECT * FROM table WHERE condition;
UPDATE table SET col = value WHERE condition;
DELETE FROM table WHERE condition;
```

Filtering

- WHERE, AND, OR, IN (...), BETWEEN ... AND ..., IS NULL, <>, LIKE
- DISTINCT

Aggregate Functions

```
COUNT(col), SUM(col), AVG(col), MIN(col), MAX(col)
GROUP BY col
HAVING condition -- filter after GROUP BY
```

Joins

```
INNER JOIN -- only matching rows
LEFT JOIN -- all from left, NULLs on right
RIGHT JOIN -- all from right, NULLs on left
ON table1.col = table2.col
```

Subqueries

```
WHERE col > (SELECT AVG(col) FROM table)
WHERE col IN (SELECT col FROM table)
WHERE col NOT IN (SELECT col FROM table)
```

Others

- UNION — combine result sets
- ORDER BY col DESC/ASC

- `LIMIT n`
- `CONCAT(col1, ' ', col2) AS alias`
- Computed columns: `(price + price * 0.18) AS price_with_tax`
- `SHOW INDEX FROM table`

Lab Progression

Lab	Topic
29	SELECT, WHERE, IN, BETWEEN
30	DISTINCT, AND/OR, ORDER BY, computed columns
31	UPDATE, DELETE, TRUNCATE
32	MIN/MAX/SUM/AVG/COUNT, LIMIT
33	UNION, CONCAT, aliases
34	GROUP BY, HAVING, IS NULL, subqueries
35	ALTER TABLE, FOREIGN KEY, DROP DATABASE
36	LEFT/RIGHT/INNER JOIN
37	Many-to-many, UNIQUE, NOT NULL
38	INDEX, SHOW INDEX

6. Java

Basics

- **Class + main:** `public class X { public static void main(String[] args) { } }`
- **Scanner:** `Scanner input = new Scanner(System.in); input.nextLine()/nextInt()/nextDouble()`
- **Printf:** `System.out.printf("%d %f %s %n", intVal, doubleVal, strVal)`
- **Type Casting:** `(double) intVal`

OOP Concepts

Concept	Keyword	Lab
Encapsulation	<code>private</code> fields + public getters/setters	Lab47
Inheritance	<code>class Child extends Parent</code>	Lab46
Abstraction	<code>abstract class X { abstract void method(); }</code>	Lab46
Interfaces	<code>interface X { void method(); } / implements</code>	Lab45
Polymorphism	Method overriding	Lab46
Constructors	Default + parameterized, overloading	Lab48
<code>this</code> keyword	Reference to current object	Lab47
<code>static</code>	Class-level variable/method, shared across instances	Lab48
<code>final</code>	Constant value, cannot be overridden	Lab48

Packages

```
package com.example;
import otherpackage.ClassName;
```

- Cross-package access with `import`

Inner Classes

- **Non-static inner:** `outer.new InnerClass()`
- **Static nested:** `static class Inner { }` — accessed as `Outer.Inner`

Arrays

```
int[] arr = {1, 2, 3};
int[][] matrix = { {1,2}, {3,4} };
for (int i : arr) { } // enhanced for-each
```

Exception Handling

```
try { riskyCode(); }
catch (Exception e) { e.printStackTrace(); }
```

File I/O

```
File file = new File("path");
file.createNewFile();
FileWriter writer = new FileWriter("path"); writer.write(data);
FileReader reader = new FileReader("path"); reader.read(charArray);
file.delete();
```

Regular Expressions

```
import java.util.regex.*;
Pattern p = Pattern.compile("\\b\\d{10}\\b");
Matcher m = p.matcher(text);
m.find() / m.matches() / m.group()
```

Date/Time API

```
LocalDateTime now = LocalDateTime.now();
DateTimeFormatter fmt = DateTimeFormatter.ofPattern("dd-MM-yyyy HH:mm:ss");
now.format(fmt);
```

Lab Progression

Lab	Topic
39	Variables, arithmetic, type casting
40	Packages, cross-package access
41	String comparison, login logic
42	1D + 2D arrays, nested loops
43	Enhanced for-each, break
44	Static nested classes, try-catch, Date/Time
45	Interfaces, multiple inheritance via interfaces
46	Abstract classes, method overriding
47	Encapsulation (private + getters/setters)
48	Constructors, static/final, inner classes
49	File I/O (create/write/read/delete)
50	Regular Expressions (Pattern/Matcher)
Exam	Pattern printing, multiplication table, ID lookup, calculator

7. Spring Boot

Project Structure (Maven)

```
src/main/java/com/example/
|-- Application.java           (@SpringBootApplication)
```

```

|-- Controller/                (@Controller, @RestController)
|-- Service/                   (@Service)
|-- Repository/                (JpaRepository interface)
|-- Models/                    (@Entity)
|-- dto/                       (Data Transfer Objects)
|-- Configurations/            (@Configuration, SecurityConfig)
'-- exception/                 (@ControllerAdvice)
src/main/resources/
  |-- application.properties
  '-- templates/                (Thymeleaf .html)
pom.xml

```

Annotations

Annotation	Purpose
@SpringBootApplication	Entry point, auto-config, component scan
@Controller	MVC controller (returns view names)
@RestController	RESTful controller (returns JSON)
@GetMapping/@PostMapping	HTTP method mapping
@RequestMapping("/path")	Class/URL-level mapping
@PathVariable	URL template variable
@RequestParam	Query string parameter
@RequestBody	JSON request body
@ModelAttribute	Form data to object binding
@Valid	Trigger validation
@Autowired	Dependency injection
@Entity	JPA entity
@Id, @GeneratedValue	Primary key
@OneToMany, @ManyToOne	Relationships
@Query	Custom JPQL
@CrossOrigin	CORS support
@ControllerAdvice	Global exception handler

Thymeleaf

```

th:text, th:utext, th:if, th:unless, th:each
th:action, th:object, th:field, th:errors
th:fragment, th:replace (layout)
@{/url} (context path), ${variable}, *{selection}

```

Spring Security

- SecurityFilterChain bean — configure HTTP security
- BCryptPasswordEncoder — password hashing
- UserDetailsService — database-backed authentication
- Form login: loginPage(), loginProcessingUrl(), defaultSuccessUrl()
- Token-based: OncePerRequestFilter, Bearer token parsing
- CSRF disable for APIs

REST API Pattern

```

POST  /api/resource  → Create
GET   /api/resource  → Read all
GET   /api/resource/{id} → Read one
PUT   /api/resource/{id} → Update
DELETE /api/resource/{id} → Delete

```


GET /api/search?keyword=... → Search

JPA Repository

```
public interface Repo extends JpaRepository<Entity, IdType> {  
    List<Entity> findByField(String value);  
    @Query("SELECT e FROM Entity e WHERE e.name LIKE %:kw%")  
    List<Entity> search(@Param("kw") String keyword);  
}
```

Additional Features

- **File Upload:** MultipartFile, Files.write(), enctype="multipart/form-data"
- **Email:** JavaMailSender, SimpleMailMessage, Mailtrap SMTP
- **PDF:** Apache PDFBox / iText
- **DTO Pattern:** decouple API from entity models
- **Validation:** @NotBlank, @Email, @Size, custom @Constraint

Lab Progression

Lab	Topic
51	First Controller, @GetMapping
52	Static HTML pages
53	Model attributes, redirect
54	Model classes, th:each
55	Thymeleaf fragments (nav/footer)
56	Forms + custom validation
57	JPA Repositories
58	JPQL custom queries
59	Full CRUD MVC
60	DTO + Service layer
61	Spring Security (form login)
62	REST APIs
63	REST Registration
64	Token-based API auth
65	File Upload + Email + PDF

8. React

Component Basics

```
function App() {  
    const [state, setState] = useState(initial);  
    useEffect(() => { }, [deps]);  
    return <div>{state}</div>;  
}  
export default App;
```

Hooks

Hook	Purpose	Lab
useState	State management	68
useEffect	Side effects on mount/update/unmount	69
useParams	URL parameters from router	70

Hook	Purpose	Lab
useNavigate	Programmatic navigation	70
useSelector	Read from Redux store	74
useDispatch	Dispatch actions to Redux store	74
useMemo	Memoized computed values	Main
useCallback	Memoized function references	Main

Props & Parent-Child

```
// Parent passes
<Child isBright={isBright} toggleLight={toggleLight} />
// Child receives
function Child({ isBright, toggleLight }) {
  return <button onClick={toggleLight}>Toggle</button>;
}
```

Conditional & List Rendering

```
{isLoggedIn ? <Dashboard /> : <Login />}
{items.map(item => <div key={item.id}>{item.name}</div>)}
```

Routing (react-router-dom)

```
const router = createBrowserRouter([
  { path: '/', element: <App /> },
  { path: 'about', element: <About /> },
  { path: 'student/:name', element: <Profile /> },
]);
<RouterProvider router={router} />
<Link to="/about">About</Link>
const navigate = useNavigate(); navigate('/login');
```

State Management

Approach	Library	Lab
Local state	useState	68
Props drilling	N/A	68
Redux Toolkit	@reduxjs/toolkit + react-redux	74
Zustand	zustand	Main

Redux Pattern:

```
const slice = createSlice({
  name: 'auth',
  initialState: { user: null },
  reducers: {
    setUser: (state, action) => { state.user = action.payload; },
    removeUser: (state) => { state.user = null; },
  }
});
```

CRUD Patterns (Local State)

```
// Create
setItems([...items, newItem]);
```

```
// Update
setItems(items.map(item => item.id === id ? { ...item, name: newName } : item));
// Delete
setItems(items.filter(item => item.id !== id));
```

REST API Integration

```
// fetch
const res = await fetch(url); const data = await res.json();
// axios
axios.post(url, data).then(res => { }).catch(err => { });
```

Protected Routes

```
function ProtectedRoute({ children }) {
  const user = useSelector(store => store.auth.user);
  if (!user) return <Navigate to="/login" replace />;
  return children;
}
```

Authentication Flow

1. User registers (POST email/password)
2. User logs in → receives token
3. Token stored in Redux + localStorage
4. Axios sends `Authorization: Bearer {token}` header
5. ProtectedRoute checks Redux store before rendering

Lab Progression

Lab	Topic
66	JSX, variables, images
67	Lists, events, conditionals
68	useState, props, child components
69	useEffect
70	React Router
71	Local CRUD
72	REST API with fetch
73	Auth with axios
74	Redux Toolkit, protected routes
75	json-server mock backend
Main	Zustand, URL shortener (Ziplink)

Quick Reference: Tech Stack Flow

HTML → Structure
 CSS/Bootstrap → Styling & Layout
 JavaScript → Client-side logic & DOM
 MySQL → Database (SQL)
 Java → Backend language (OOP)
 Spring Boot → Framework (MVC, REST, Security, JPA)
 React → Frontend framework (SPA)

Full Stack Flow:

React (UI) → Spring Boot REST API (Backend) → JPA Repository → MySQL Database
=>
Spring Security (Auth + Tokens)